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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0002 - WEU		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2024-10-29
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	1		16			
2	X	1,5		17			
3	X	3		18			
4	X	3		19			
5	X	1		20			
6	X	1,5		21			
7	X	2,5		22			
8	X	1,5		23			
9	X	6		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		21	Betyg / Grade: E	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
10	0	OK	F.W.

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-10-28 kl. 12:37:16

Example Solutions for DT081A – Signal and Image Processing Exam, 2024-10-29

1 (1 + 1 + 1 + 1 + 1 = 5 p)

a)
The process of converting a continuous-time signal into a discrete-time signal by taking periodic samples, $x[n] = x(nT_s)$ where T_s is the sampling interval.

b)
The process of mapping a continuous range of values to a finite set of discrete levels, often in analog-to-digital conversion.

c)
The output of a system when an impulse signal is applied at its input. It characterizes the system's behavior.

d)
The steady-state response of a system to sinusoidal inputs at varying frequencies.

e)
A technique used to reduce spectral leakage in signal analysis by applying a window function to a signal before performing a Fourier transform.

2 (3 + 2 = 5 p)

a)
$y[n] = s[n] * h[n]$

$y[0] = (3)(-0.5) = -1.5$
$y[1] = (7)(-0.5) + (3)(0) = -3.5$
$y[2] = (10)(-0.5) + (7)(0) + (3)(0.5) = -5 + 1.5 = \mathbf{-3.5}$
$y[3] = (3)(-0.5) + (10)(0) + (7)(0.5) = -1.5 + 3.5 = \mathbf{2}$
...

$y[n] = \{-1.5, -3.5, -3.5, 2, 7.5, 1.5, -2.5\}$

b) This filter approximates a differentiation operation, emphasizing changes in the signal, which is characteristic of a high-pass filter.

3 (1 + 2 = 3 p)

a) The system diagram shows both a feed-forward and feedback branch. The latter implies that the system will produce an infinite impulse response IIR.

b)
$y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$

4 (2 + 3 = 5 p)

a)
Extend spectrum plot to at least $-2\pi < \Omega < 2\pi$, where the continous spectrum is mirrored on the y-axis and replicated over the whole Ω axis, with a periodicity of 2π.

b)
$f_s \geq 2f_{max}$
The lowest sampling frequency depends on the signal bandwidth. The figure shows that $f_{max} = \frac{3}{4}\pi\frac{f_s}{2\pi} = \frac{3}{8}f_s = \frac{150}{8} = 18.75Hz$.
Hence $f_s \geq 37.5Hz$.

5 (2 + 4 = 6 p)

a) Amplitude and phase relationship:
$|Y(f)| = |H(f)||X(f)|$
$\angle Y(f) = \angle H(f) + \angle X(f)$

b) Sketch spectra

$$|Y(f)| = \begin{cases} 8f & \text{for } 0 \leq f \leq 10 \\ 80 - 4(f-10) & \text{for } 10 < f \leq 30 \\ 0 & \text{for } f < 0 \text{ or } f > 30 \end{cases}$$

and

$arg(Y(f)) = 0 \forall f$

6 (2 + 2 = 4 p)

a) Window iii) has the narrowest main lobe, which provides the best frequency separation between closely spaced frequency components allowing for better distinction.
b) Window iv) has zero outside of main lobe, which effectively eliminates spectral leakage in the frequency domain ensuring that frequency components are not contaminated by artifacts from distant frequencies.

7 (3 p)

At low frequencies the system has an amplification of 1, whereas higher frequencies are amplified more. Combined this results in high frequency amplification (not to be confused with a high pass filter), which if applied to an image would enhance edges.

8 (1 + 1 + 1 + 1 + 1 = 5 p)

- $1 \rightarrow ii$
  Only the center element is used, and reduce the input with 0.5.
- $2 \rightarrow iv$
  A vertical filter emphasized vertical edges only.
- $3 \rightarrow v$
  Edge detecting kernel with elements summing to 0, only shows edges.
- $4 \rightarrow iii$
  Avering kernel smooths the image as a low-pass filter.
- $5 \rightarrow i$
  Edge detecting kernel summed with a unit kernel translates kernel to one with center element = 5, with elements summing to 1.

9 (6 p)

$x(i,j)$ has several non-zero values, that each contribute to the $y(i,j)$ through convolution with $h(i,j)$. From examining the isolated scaled impulse responses we can determine that $h(i,j)$ has a cross-shaped pattern. By subtracting one scaled impulse reponse from another we can identify the impulse response as.

$$h(i,j) = \begin{bmatrix} 0 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 0 \end{bmatrix}$$

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Lärarens anteckning / Teachers note:

1) a) Sampling means extracting a signal in a certain frequency (usually at 2 times the original signals max frequency) in order to recreate it in a new signal. 0,5

b) Quantization is a term used for Signal Quantization Noise Ratio (SQNR) in order to decide the number of bits used for an image to reduce noise in the image. 0

c) The impulse response is the effect of a system when an input (impulse) is given to a system which results in a specific output. $x(n) \rightarrow H \rightarrow y(n)$ (an impulse response; H: system) 0

d) The frequency response is the effect of an images frequency when applying things like a filter (low-pass, high-pass etc). The effect can then be seen in a graph or spatial domain to study high (sharp edges) and low (remove edges gives a blurry effect) frequencies in an image. 0,5

1

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

2)a) $s(n) = \{3, 7, 10, 3, -5\}$

$h(n) = \{-0.5, 0, 0.5\}$

$$y(n) = \sum_{k=0}^{4} h(k) \cdot s(n-k)$$

$y(0) = h(0) \cdot s(0) = -0.5 \cdot 3 = -1.5$

$y(1) = s(1) \cdot h(0) + h(1) \cdot s(0) = 7 \cdot -0.5 + 0 \cdot 3 = -3.5$

$y(2) = s(0) \cdot h(2) + s(1) \cdot h(1) + s(2) \cdot h(0) = 3 \cdot 0.5 + 7 \cdot 0 + 10 \cdot -0.5 = -3.5$

$y(3) = s(3) \cdot h(0) + s(2) \cdot h(1) + s(1) \cdot h(2) = 3 \cdot -0.5 + 10 \cdot 0 + 7 \cdot 0.5 = 2$

$y(4) = s(4) \cdot h(0) + s(3) \cdot h(1) + s(2) \cdot h(0) = -5 \cdot -0.5 + 3 \cdot 0 + 10 \cdot -0.5 = -2.5$

1,5

$y(n) = \{-1.5, -3.5, -3.5, 2, -2.5\}$ $y(5) - y(7)$

b) The resulting output shows a dampening of the signal making me believe its an attenuation of the signal using the filter.

0

1,5

Lärarens anteckning / Teachers note:

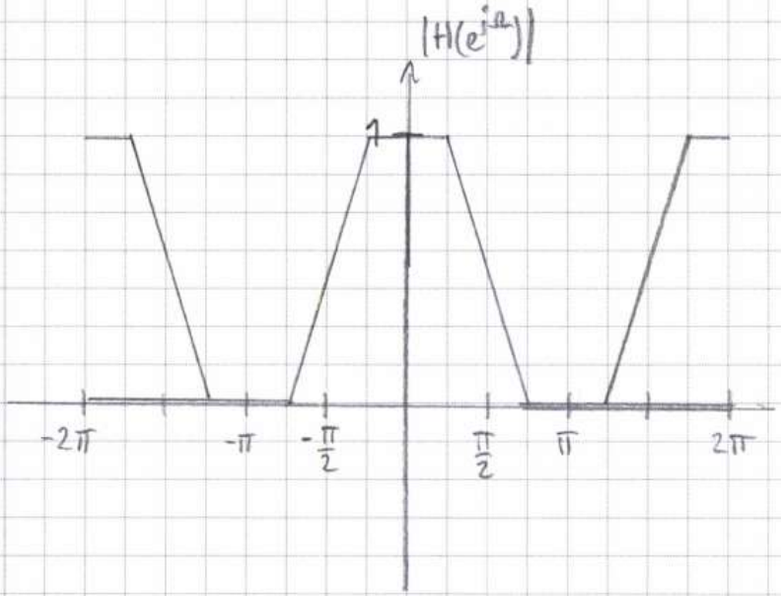
3)a) This is a Infinite Impulse Response (IIR) filter since it takes an input signal $x(n)$ and then has a feedback of the signal from $y(n)$ back to itself using delays and weights.
The system will go on forever since the output signal becomes the input signal of the system. 1

b) $y(n) = x(n) + 2x(n-1) + x(n-3) - 3y(n-1) + 2y(n-3)$
Above is the difference equation of the system. 2

3

Lärarens anteckning / Teachers note:

4)a)



2

b) According to Nyquist sampling theorem the lowest sampling frequency required in order to reconstruct the signal correctly is $f_s \geq 2 \cdot f_{max}$

The period of the frequency: $T = \frac{1}{f}$ =

$T = 2\pi$, $f = \frac{1}{2\pi}$

$f_s \geq 2 \cdot \frac{1}{2\pi} = \frac{1}{\pi}$

1

Answer: lowest sampling frequency $f_s = \frac{1}{\pi}$

3

Lärarens anteckning / Teachers note:

5)a)

$x(t) \rightarrow H(f) \rightarrow y(t)$

$|Y(f)| = |H(f)| \cdot |X(f)|$

where $Y(f)$ is the output, $H(f)$ is the amplitude and $X(f)$ is the input.

The phase of the amplitude:

$$H(e^{j\Omega}) = |H(e^{j\Omega})| e^{j\arg H(e^{j\Omega})}$$

where $e^{j\arg H(e^{j\Omega})}$ is the phase

$$e^{j\arg H(e^{j\Omega})} = H \cdot \cos(\Omega) + H \cdot j \cdot \sin(\Omega)$$

Re Im

1

1

Lärarens anteckning / Teachers note:

6)a) The window with the best frequency separation is iii since it has both the high frequency of sharp edges and low frequencies showing different tones of color (or shades of gray). The lower ripples are more frequent compared to window i which indicates more color tones. 1

b) Assuming a high pass filter has been used where only the high frequency (sharp edges) remain, then the best window would be iv since it has removed the low frequencies and only the high frequency remain. 0,5

1,5

7) The system clearly shows that the amplitude of the center of the two-dimensional image is 0. Which indicates that the lower frequencies have been attenuated and only the higher frequencies on the edges of the figure have higher amplitude which shows that the system only shows the edges of the image which have high frequencies.

Lärarens anteckning / Teachers note:

2,5

Lärarens anteckning / Teachers note:

8) 4 with iii , the image is blurry which is a result of multiplying $\frac{1}{9}$ to the kernel and decreasing the sharpness of the image. 1

2 with ii , the image has gone darker and a little out of focus which happens with negative values on the side of the kernel together with the column of zeroes which removes the focus of the image. 0

5 with v , the negative values in the kernel give a darker look removing the low frequency values together with the zeroes in the corners. The higher value in the middle increase the sharpness which leaves a higher frequency and results in an image showing all the edges in the image. 0,5

3 with iv , same as the above but with a lower value on the center value in the kernel which gives less sharper edges (higher frequency). 0

1 with i , The image has a lot of noise which is the result of reducing the pixel values by 0,5 and removing pixel values around with the zeroes in the kernel. 0

1,5

Lärarens anteckning / Teachers note:

9)

$$h(i,j) = \begin{bmatrix} 0 & 0{,}125 & 0 \\ 0{,}125 & 0{,}5 & 0{,}125 \\ 0 & 0{,}125 & 0 \end{bmatrix}$$

$X(i,j)$:

i \ j	0	1	2	3	4	5
0	0	0	0	0	0	144
1	0	112	0	0	0	0
2	0	0	0	224	0	0
3	80	0	64	0	0	0
4	0	0	0	176	0	0
5	0	0	0	0	0	48

$y(0,0) = 0 \cdot 112 = 0$, $y(0,1) = 0{,}125 \cdot 112 = 14$, $y(0,2) = 0 \cdot 112$

$y(1,0) = 0{,}125 \cdot 112 = 14$, $y(1,1) = 0{,}5 \cdot 112 = 56$, $y(1,2) = 0{,}125 \cdot 112 = 14$

$y(2,0) = 0 \cdot 112 = 0$, $y(2,1) = 0{,}125 \cdot 112 = 14$, $y(2,2) = 0 \cdot 112 = 0$

$y(1,2) = 0 \cdot 224 = 0$, $y(1,3) = 0{,}125 \cdot 224 = 28$, $y(1,4) = 0 \cdot 224 = 0$

$y(2,2) = 0{,}125 \cdot 224 = 28$, $y(2,3) = 0{,}5 \cdot 224 = 112$, $y(2,4) = 0{,}125 \cdot 224 = 28$

$y(3,2) = 0 \cdot 224 = 0$, $y(3,3) = 0{,}125 \cdot 224 = 28$, $y(3,4) = 0 \cdot 224 = 0$

$y(0,4) = 0{,}125 \cdot 144 = 18$, $y(0,5) = 0{,}5 \cdot 144 = 72$, $y(1,4) = 0 \cdot 144 = 0$

$y(1,5) = 0{,}125 \cdot 144 = 18$

$y(2,0) = 0{,}125 \cdot 80 = 10$, $y(2,1) = 0 \cdot 80 = 0$, $y(3,0) = 0{,}5 \cdot 80 = 40$, $y(3,1) = 10$

$y(4,0) = 10$, $y(4,1) = 0$

$y(2,1) = 0 \cdot 64 + 14 = 14$, $y(2,2) = 0{,}125 \cdot 64 = 8 + 28 = 36$, $y(2,3) = 0 \cdot 64 + 112 = 112$

$y(3,1) = 0{,}125 \cdot 64 + 10 = 18$, $y(3,2) = 0{,}5 \cdot 64 = 32$, $y(3,3) = 0{,}125 \cdot 64 + 28 = 36$

$y(4,1) = 0 \cdot 64 = 0$, $y(4,2) = 0{,}125 \cdot 64 = 8$, $y(4,3) = 0 \cdot 64 = 0$

$y(3,2) = 0 \cdot 176 = 0$, $y(3,3) = 0{,}125 \cdot 176 + 36 = 58$, $y(3,4) = 0 \cdot 176 = 0$

Lärarens anteckning / Teachers note:

9) $y(4,2) = 0{,}125 \cdot 176 + 8 = 30$, $y(4,3) = 0{,}5 \cdot 176 = 88$

$y(4,4) = 0{,}125 \cdot 176 = 22$

$y(5,2) = 0 \cdot 176 = 0$, $y(5,3) = 0{,}125 \cdot 176 = 22$, $y(5,4) = 0 \cdot 176 = 0$

$y(4,4) = 0 \cdot 48 + 22 = 22$, $y(4,5) = 0{,}125 \cdot 48 = 6$

$y(5,4) = 0{,}125 \cdot 48 = 6$, $y(5,5) = 0{,}5 \cdot 48 = 24$

Sum all these up and place zero in the ones the kernel doesnt affect.

$y(i,j) =$

i \ j	0	1	2	3	4	5
0	0	14	0	0	18	72
1	14	56	14	28	0	18
2	10	14	36	112	28	0
3	40	18	32	58	0	0
4	10	0	30	88	22	6
5	0	0	0	22	6	24

The kernel transforms the input image by placing it centered around the box with a value in the input image and then multiplying the surrounding boxes with its corresponding boxes in the kernel. If two boxes overlap you perform addition on the values in the box.

/6